Title: AI-Native Multi-Agent System for Micro, Small, and Medium Enterprises

Submitted By:
Sanskar Awati [BEAIDS22615],
Moinuddin Ansari [BEAIDS22619],
Ankush Dhakare [BEAIDS22629],
Abhishek Kute [BEAIDS22645]
Guided by:
Mrs. Reena Sahane
Ms. Surbhi Pagar

Problem Statement:

Micro, Small, and Medium Enterprises (MSMEs) struggle to compete with large corporations that utilize advanced data analytics and automation to dominate markets and optimize supply chains. Often lack the tools to make data-driven decisions and manage vendor relations effectively, which frequently leads to the detrimental Bullwhip effect. Our economical AI-Native Multi-Agent solution directly addresses this by providing MSMEs with a comprehensive system for data analytics, visualization, forecasting, and vendor management, empowering them to compete on equal footing in a technology-driven market.

Abstract:

This project introduces an AI-Native Multi-Agent System built to improve decision-making for Micro, Small, and Medium Enterprises. It combines internal data with external signals such as market trends and social media activity. The system uses a secure JWT-based architecture supported by specialized agents. The Data Harvester collects input from formats like CSV files or Google Sheets. The Trend Analyst extracts the trends from social media and current market situation, geographically. The Forecaster improves predictions by using a better Prophet model with extra parameters. Using Monte Carlo Tree Search (MCTS), the system analyzes countless reordering plans to optimize inventory levels and avoid both stockouts and excess stock. The Order Manager automates purchasing, while the Visualizer presents real-time analytics. All agents operate under a state-driven control loop, allowing for adaptive and autonomous decisions. The solution reduces manual work, improves forecasting, and gives MSMEs a scalable AI-driven infrastructure that supports competitive growth in a technology-focused market.

Keywords: AI Models, Prophet Model, Monte Carlo Tree Search, AI Agents, Sales Trend Analysis, Visualization

Introduction:

Micro, Small, and Medium Enterprises are an integral component of the majority of economies but are not always able to compete with the resources and infrastructures of the large corporation. Lacking sophisticated forecasting, supply chain optimization, and vendor management tools, they are operationally inefficient and also more at risk. A significant supply chain problem is the Bullwhip effect, wherein a small change in customers' purchases triggers much larger oscillations in orders and inventory as it goes through suppliers.

With a view to solving these challenges, we have created the AI-Native Multi-Agent System. This system employs synchronized AI agents to run data analysis, prediction, vendor communication, and purchasing in one system. Through automation and transparent insight, it enables MSMEs to make effective decisions without requiring large technical teams. The vision is to make large-scale technological capabilities affordable to smaller enterprises and enable them to compete on par.

Motivation:

- MSMEs account for over 90 percent of world firms but continue to have no access to sophisticated decision-making tools.
- Manual processes in procurement, inventory, and sales cause delays and loss of revenue.
- Such demand is increasing for simple-to-use, interpretable, and low-cost AI-based systems that suit the specific requirements of MSMEs.
- Minimizing the Bullwhip effect stabilizes operations, leading to improved efficiency as well as more robust profit margins.

System Architecture:

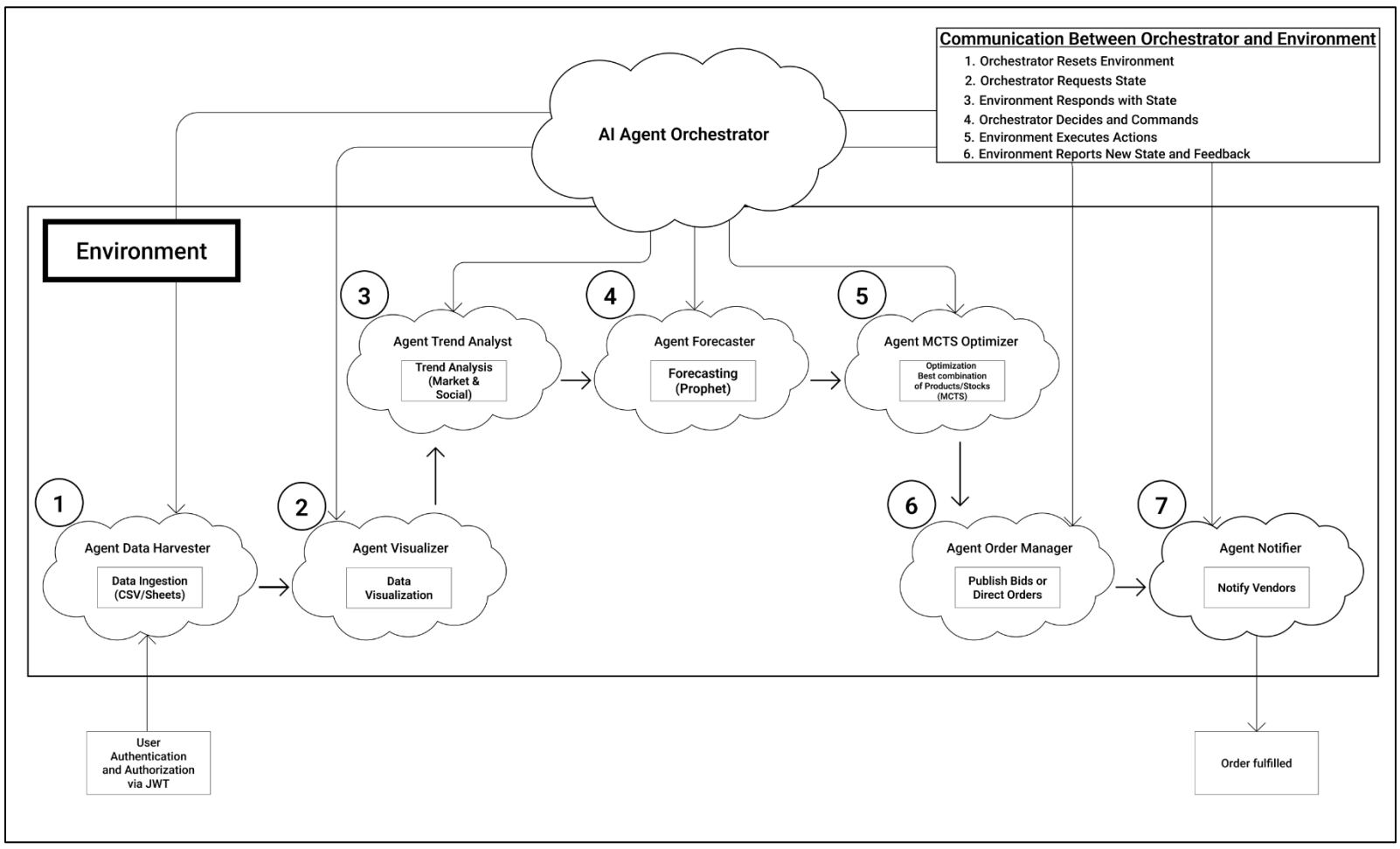

Figure 1. AI-Native Multi-Agent Architecture

Technical Details:

Problem Description:

While large companies are utilizing powerful analytics and automation systems, small businesses are lagging behind due to lack of budget and skills. With that lack of access, small firms cannot do demand forecasting, optimize supply chains or engage in effective vendor communications. Compounding this effect is the Bullwhip effect, where small fluctuations in demand are amplified to further fluctuations to the supply chain, leading to a breakdown in expectations, costly purchases, wasteful returns, and poor visibility / management of stock. Our project is focused on providing an integrated solution at the low cost, accessible, with lower skill use to alleviate some of these burdens.

Research Questions:

1. In what way can automated vendor management and forecasting bidding lessen the Bullwhip effect in MSME supply chains?
2. How does Monte Carlo Tree Search facilitate decision making in dynamic purchasing and inventory scenarios?
3. Is there an autonomous AI-Native Multi-Agent system that can obtain better automation and / or better decision-making quality over traditional rule-based systems, for MSMEs?
4. What are the pros and cons of a centralized AI orchestrator that manages autonomous agents, particularly in terms of accuracy, scalability, and trust by non-technical users?

Objectives:

- Create a robust, modular, and scalable multi-agent system for MSMEs that can automate procurement, forecasting, and vendor management.
- Adopt time-series forecasting models like Prophet with additional inputs to enhance predictions in low-data and high-variability settings.
- Minimize the Bullwhip effect by automatic order placement and forecasted bidding by analyzing live data.
- Develop a core AI Orchestrator that enlists all agents with decision-making remaining explainable and simple to comprehend for non-technical users.

Potential Outcome:

- **Timely better decisions:** Integrating the Data Harvester, Visualizer, and Trend Analyst will give MSMEs the ability to track key metrics and better respond to changes in the market in real time.
- **Ability to forecast with greater accuracy:** Using a mix of market and public data will increase MSME's ability forecast evenings sales and bidding strategy.
- **Computer-automated supply chain optimization:** Order Manager and MCTS Optimizer will adjust bidding and order automatically to minimize stock outs and surplus inventory.
- **Scalable and Modular:** The AI-Native Multi-Agent Architecture's modularity enables to develop and grow single agents (i.e., Enhanced JWT, Enhanced Data Pipeline) separately and thus allows system scalability and future-proofing for new market demands.

Applications:

- **Market Trend Analysis:** The Agent Trend Analyst and Visualizer would allow firms to analyze trends from public social media and market data, and quickly change their products to align with the trends consumers are following.
- **Financial Planning and Forecasting:** The Prophet would forecast sales, revenue, and spending to assist MSMEs with planning financials, budgeting, and risk.
- **Strategic Procurement and Bidding:** The MCTS Optimiser and Order Manager would automatically be able to prepare and submit bids for the MSME to ensure that the MSMS procure at the optimum terms with volatility in the market, and prevent the MSME from missing out on bidding situations.
- **Supply Chain Optimisation:** MSMEs would use the system to automate the supply chain demand forecasting, ordering, and inventory control process preventing unnecessary surplus stock, while eliminating running out of stock.